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CSCI 347 Artificial Intelligence
**Project 1: A* Search in Block Maze Write Up**
**16 February 2026**

## *Definition of Search Problem*

The rolling die maze problem can be formalized as a search problem with the following components:

- All States consist of three components:
  - Position: The row and column coordinates
  - Die Orientation: Belonging to a Dice object, where all faces correspond to cardinal directions. The die's remaining faces (bottom, south, west) are determined by the constraint that opposite faces sum to 7 (1↔6, 2↔5, 3↔4).

  We represent this as a tuple: (start_row, start_col, Dice(top=1, north=2, east=3))
  The **initial state** specifically has row and column coordinates of the start square (S) and begins with face 1 on top, face 2 facing north, and face 4 facing east.

- Goal Test: a state satisfies the goal test if and only if:
  1. The die's position matches the goal location G, AND
  2. The die has face 1 on top

  Both conditions must be met simultaneously. Simply reaching position G with any other face on top does not satisfy the goal.

- Actions: from any state, the agent can attempt four cardinal direction movements:
  - North: Roll the die forward (decreasing row coordinate)
  - South: Roll the die backward (increasing row coordinate)
  - East: Roll the die right (increasing column coordinate)
  - West: Roll the die left (decreasing column coordinate)

  Each roll physically changes the die's orientation. For example, rolling north causes the top face to become the new north face, and the south face to become the new top face.

- State Constraints: Not all actions are valid from every state. An action is only applicable if:
  - The resulting position is within the maze boundaries
  - The resulting position is not an obstacle (marked with *)
  - After rolling, face 6 is not on top (critical constraint)

- Path Cost: This problem uses uniform cost, where each valid roll action has a true G cost of 1. The path cost is therefore the total number of moves (rolls) from the initial state to the goal state.

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## *Formal Definitions of Heuristics*

We implemented two admissible heuristics for A* search.

**Heuristic 1: Manhattan Distance**
Formal Definition:
*h(n) = |current row - goal row| + |current col - goal col|*

This heuristic measures the minimum number of moves required to reach the goal position if the die could move freely through obstacles and without orientation constraints. It represents the straight-line grid distance.

Manhattan distance is admissible because it never overestimates the true cost to reach the goal. The die cannot reach the goal in fewer moves than the Manhattan distance, since each roll moves exactly one square in a cardinal direction. The heuristic also ignores obstacles, which realistically may require detours, AND die orientation constraints, which may require additional moves, making it optimistic and admissible.

**Heuristic 2: Manhattan Distance with Orientation Penalty (Reorientation)**
Formal Definition:
*h(n) = |current row - goal row| + |current col - goal col| + penalty*
where *penalty = 0 if top face = 1,  +1 if top_face ≠ 1*

This heuristic extends Manhattan distance by accounting for die orientation. It recognizes that if the die does not currently have face 1 on top, additional moves will be required to satisfy the goal test.

This heuristic is admissible because the penalty of 1 never overestimates the true reorientation cost. If the die needs to have face 1 on top at the goal but currently has a different face on top, at least one additional roll is required to change the top face. In practice, reorientation often requires 2 or more moves depending on the current orientation, but we use a penalty of only 1 to maintain admissibility.

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## *Results*

We tested both heuristics on five maze configurations of varying complexity. The results are shown in the table below.

**Manhattan Heuristic**

| **Maze #** | **1** | **2** | **3** | **4** | **5** |
|---|---|---|---|---|---|
| **Path Length** | 6 | 16 | N/A | 21 | 26 |
| **Nodes Gen.** | 22 | 77 | 3 | 128 | 1126 |
| **Nodes Exp.** | 15 | 57 | 3 | 101 | 680 |

**Reorientation Heuristic**

| **Maze #** | **1** | **2** | **3** | **4** | **5** |
|---|---|---|---|---|---|
| **Path Length** | 6 | 16 | N/A | 21 | 26 |
| **Nodes Gen.** | 21 | 77 | 3 | 115 | 1153 |
| **Nodes Exp.** | 14 | 57 | 3 | 89 | 680 |

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## *Discussion and Analysis*

The results seem to be consistent with A* search theory. We say that the reorientation heuristic is strictly better than the Manhattan heuristic, but only on a **local** scale because it only adds information at the very final square of the maze. For example, reorientation heuristics helped in Maze 4 because the search avoids expanding dead-ends that look like solutions based on position and not orientation. However, it didn't help in Maze 5 because it generated more nodes than the Manhattan distance heuristics and this could have been due to tie-breaking and the algorithm's unnecessary prioritization of coordinates with 1 on top that aren't actually goal states. In other words, the reorientation heuristics improves the performance on problems where the path is misleading to the wrong orientation goal and performs equally to the Manhattan heuristic when the problem state is just getting to the goal.

Since both heuristics do not account for obstacles, it was assumed that they would perform quite similarly between the different mazes. This is why our results were interesting, since even such a slight penalty can make a significant difference in the amount of nodes generated and expanded. Should we have picked a heuristic that accounts for obstacles and one that does not, our results might have been drastically different.